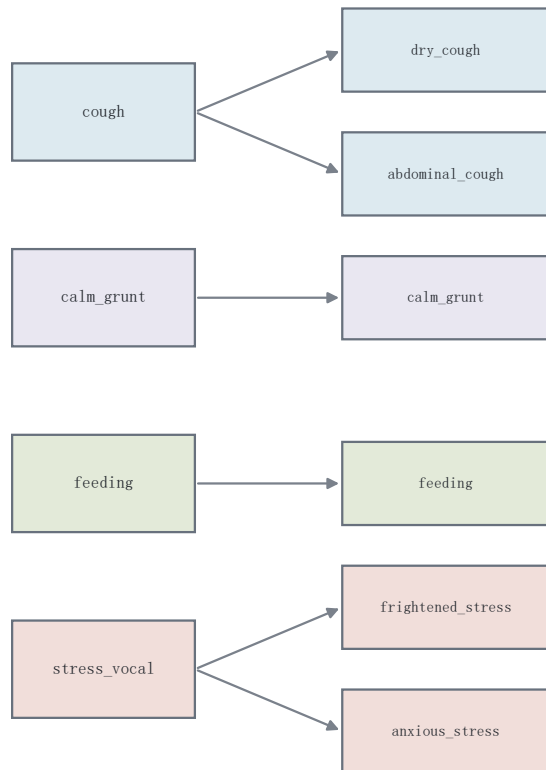


a

Main classes

Auxiliary subtypes

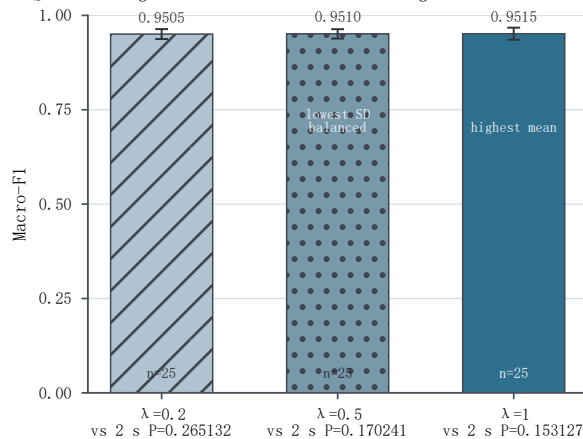


Auxiliary supervision is used during training; the main task remains four-class classification.

b

Auxiliary-loss weight

No significant incremental clean gain



c

Clean post-hoc prototype inference

Same $\lambda=0.5$ cohort; no significant paired gain